2.7.28

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Question

Find the area of the triangle whose vertices are $A(2,3),\ B(3,5),\ C(4,4)$.

The vertices of a triangle

Table: Vectors

The sides of the triangle

$$(\mathbf{A} - \mathbf{B}) = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \tag{1}$$

$$(\mathbf{A} - \mathbf{C}) = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix}. \tag{2}$$

Cross product

Using (**??**) and (**??**) The magnitude of the cross product is

$$\|(\mathbf{A}-\mathbf{B})\times(\mathbf{A}-\mathbf{C})\| = \sqrt{0^2+0^2+(-3)^2} = 3. \tag{3}$$

Area of triangle

Therefore the area of triangle ABC is

$$\text{ar}(\triangle ABC) = \frac{1}{2}\left\|(\mathbf{A}-\mathbf{B})\times(\mathbf{A}-\mathbf{C})\right\| = \frac{1}{2}\times 3 = 1.5. \tag{4}$$

C Code

```c
#include <stdio.h>
#include <math.h>

// Function to compute cross product magnitude in 3D
double area_triangle(double A[3], double B[3], double C[3]) {
    double AB[3] = {A[0] - B[0], A[1] - B[1], 0};
    double AC[3] = {A[0] - C[0], A[1] - C[1], 0};
    double cross[3] = {
        AB[1]*AC[2] - AB[2]*AC[1],
        AB[2]*AC[0] - AB[0]*AC[2],
        AB[0]*AC[1] - AB[1]*AC[0]
    };
    double mag = sqrt(cross[0]*cross[0] + cross[1]*cross[1] +
        cross[2]*cross[2]);
    return 0.5 * mag;
}
```

C Code - Main

```c
int main() {
    double A[3] = {2,3,0};
    double B[3] = {3,5,0};
    double C[3] = {4,4,0};

    double area = area_triangle(A,B,C);
    printf("Area of triangle = %.2f\n", area);

    FILE *fp = fopen("triangle_points.txt","w");
    fprintf(fp,"%f %f\n", A[0], A[1]);
    fprintf(fp,"%f %f\n", B[0], B[1]);
    fprintf(fp,"%f %f\n", C[0], C[1]);
    fprintf(fp,"%f %f\n", A[0], A[1]); // close loop
    fclose(fp);

    return 0;
}
```

Python + C Code

```python
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load compiled C library
# Compile C code first: gcc -shared -o triangle_area.so -fPIC
    your_c_code.c
lib = ctypes.CDLL("./triangle_area.so")

# Define argument and return types
lib.area_triangle.argtypes = [ctypes.POINTER(ctypes.c_double),
                              ctypes.POINTER(ctypes.c_double),
                              ctypes.POINTER(ctypes.c_double)]
lib.area_triangle.restype = ctypes.c_double
```

```python
# Define points
A = (ctypes.c_double * 3)(2,3,0)
B = (ctypes.c_double * 3)(3,5,0)
C = (ctypes.c_double * 3)(4,4,0)

# Call C function
area = lib.area_triangle(A,B,C)
print("Area of triangle =", area)

# Plot triangle
points = np.array([[2,3],[3,5],[4,4],[2,3]])
plt.plot(points[:,0], points[:,1], 'bo-')
for i, txt in enumerate(['A','B','C']):
    plt.text(points[i,0]+0.1, points[i,1]+0.1, txt, fontsize=12)
plt.title(f"Triangle ABC (Area = {area:.2f})")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.axis("equal")
```

Native Python Code

```python
import numpy as np
import matplotlib.pyplot as plt

# Define points
A = np.array([2,3,0])
B = np.array([3,5,0])
C = np.array([4,4,0])

# Vectors
AB = A - B
AC = A - C

# Cross product & area
cross = np.cross(AB,AC)
area = 0.5 * np.linalg.norm(cross)
print("Area of triangle =", area)
```

```python
# Plot triangle
points = np.array([[2,3],[3,5],[4,4],[2,3]])
plt.plot(points[:,0], points[:,1], 'ro-')
for i, txt in enumerate(['A','B','C']):
    plt.text(points[i,0]+0.1, points[i,1]+0.1, txt, fontsize=12)
plt.title(f"Triangle ABC (Area = {area:.2f})")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.axis("equal")
plt.show()
```

beamer/figs/Figure_1.png